

Regression Assumptions After Modeling

Executive summary report for the New York City Taxi and Limousine Commission

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ISSUE / PROBLEM

The New York City Taxi and Limousine Commission wanted a way to predict taxi fares before a ride starts. The team needed to build a regression model that calculates accurate fare amounts for passengers in advance.

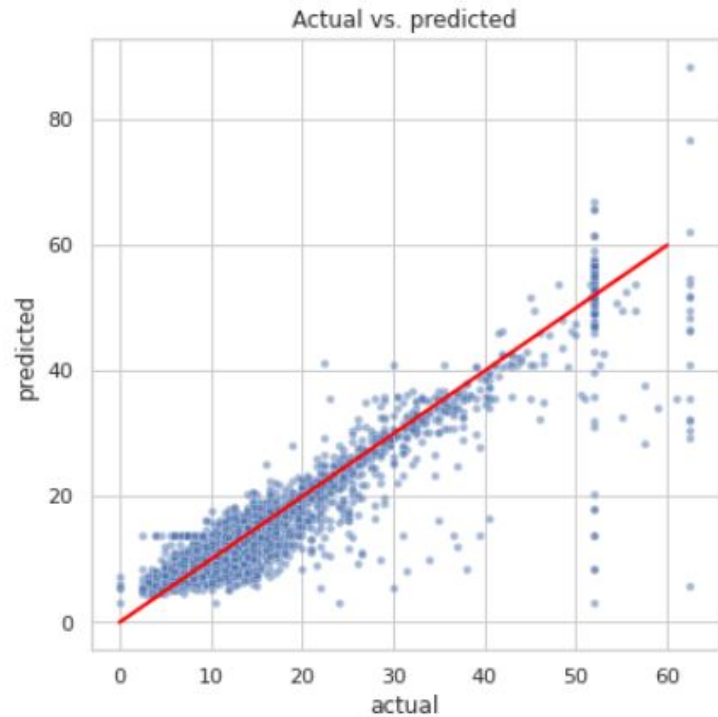
RESPONSE

- We cleaned the data by fixing bad numbers and removing extreme outliers.
- We created a new feature called **mean_distance** to estimate trip distance before the ride begins.

IMPACT

Cleaning outliers and choosing the right features made the model very reliable. The city now has a working model to give riders accurate price quotes before they book a taxi.

The data team evaluated the model using a scatter plot comparing actual fare costs to our predictions. This confirmed that the model provides dependable fare estimates. Full residual diagnostic tests can be reviewed in the accompanying notebook.



Model metrics:

- Net model tuning resulted in:
 - ✓ R^2 0.87, meaning that 86.8% of the variance is described by the model.
 - ✓ MAE 2.1
 - ✓ MSE: 14.36
 - ✓ RMSE 3.8

KEY INSIGHTS

- Trip distance is the biggest factor in determining the fare. Time of day has very little effect.
- Because fares rely heavily on trip distance, the city can easily estimate prices before a trip starts
- The NYC Taxi and Limousine Commission can use these findings to build an upfront fare calculator into a passenger app. This gives riders clear price estimates before booking.
- Drivers can also use these distance-based estimates to better anticipate earnings for longer routes before accepting a trip.